Write your answers in the D1 answer book for this paper.

1. The table below shows the distances, in km, between six data collection points, A, B, C, D, E and F.

	A	**B**	**C**	**D**	**E**	**F**
A	–	35	42	55	48	50
B	35	–	40	49	52	31
C	42	40	–	47	53	49
D	55	49	47	–	39	44
E	48	52	53	39	–	52
F	50	31	49	44	52	–

Ferhana must visit each data collection point. She will start and finish at A and wishes to minimise the total distance she travels.

(a) Starting at A, use the nearest neighbour algorithm to obtain an upper bound for the distance Ferhana must travel. Make your method clear.

(2)

(b) Starting by deleting B, and all of its arcs, find a lower bound for the distance Ferhana must travel. Make your calculation clear.

(3)

(Total 5 marks)

Leave blank

1.

	A	B	C	D	E	F
A	–	35	42	55	48	50
B	35	–	40	49	52	31
C	42	40	–	47	53	49
D	55	49	47	–	39	44
E	48	52	53	39	–	52
F	50	31	49	44	52	–

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Question 1 continued

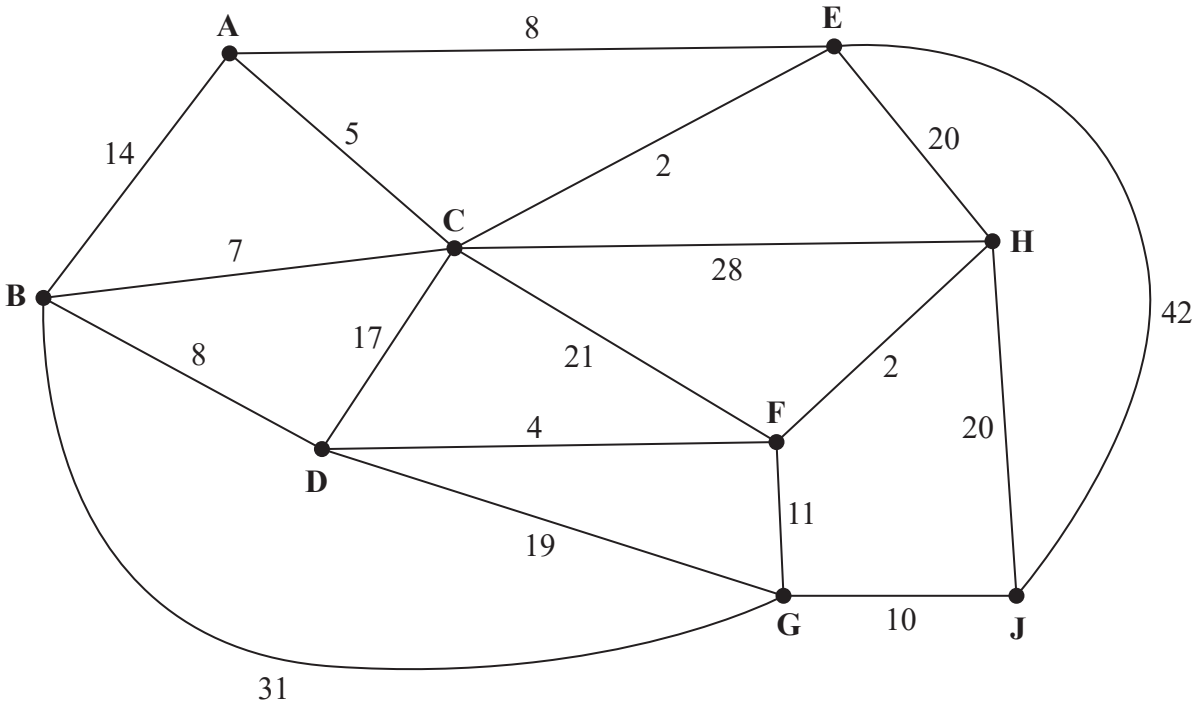
	A	B	C	D	E	F
A	–	35	42	55	48	50
B	35	–	40	49	52	31
C	42	40	–	47	53	49
D	55	49	47	–	39	44
E	48	52	53	39	–	52
F	50	31	49	44	52	–

Q1

(Total 5 marks)

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6.



[*The total weight of the network is 269*]

Figure 3

Figure 3 models a network of roads. The number on each edge gives the time taken, in minutes, to travel along the corresponding road.

(a) Use Dijkstra's algorithm to find the shortest time needed to travel from A to J. State the quickest route.

(6)

Alan needs to travel along all the roads to check that they are in good repair. He wishes to complete his route as quickly as possible and will start at his home, H, and finish at his workplace, D.

(b) By considering the pairings of all relevant nodes, find the arcs that will need to be traversed twice in Alan's inspection route from H to D. You must make your method and working clear.

(5)

For Alan's inspection route from H to D

(c) (i) state the number of times vertex C will appear,

(ii) state the number of times vertex D will appear.

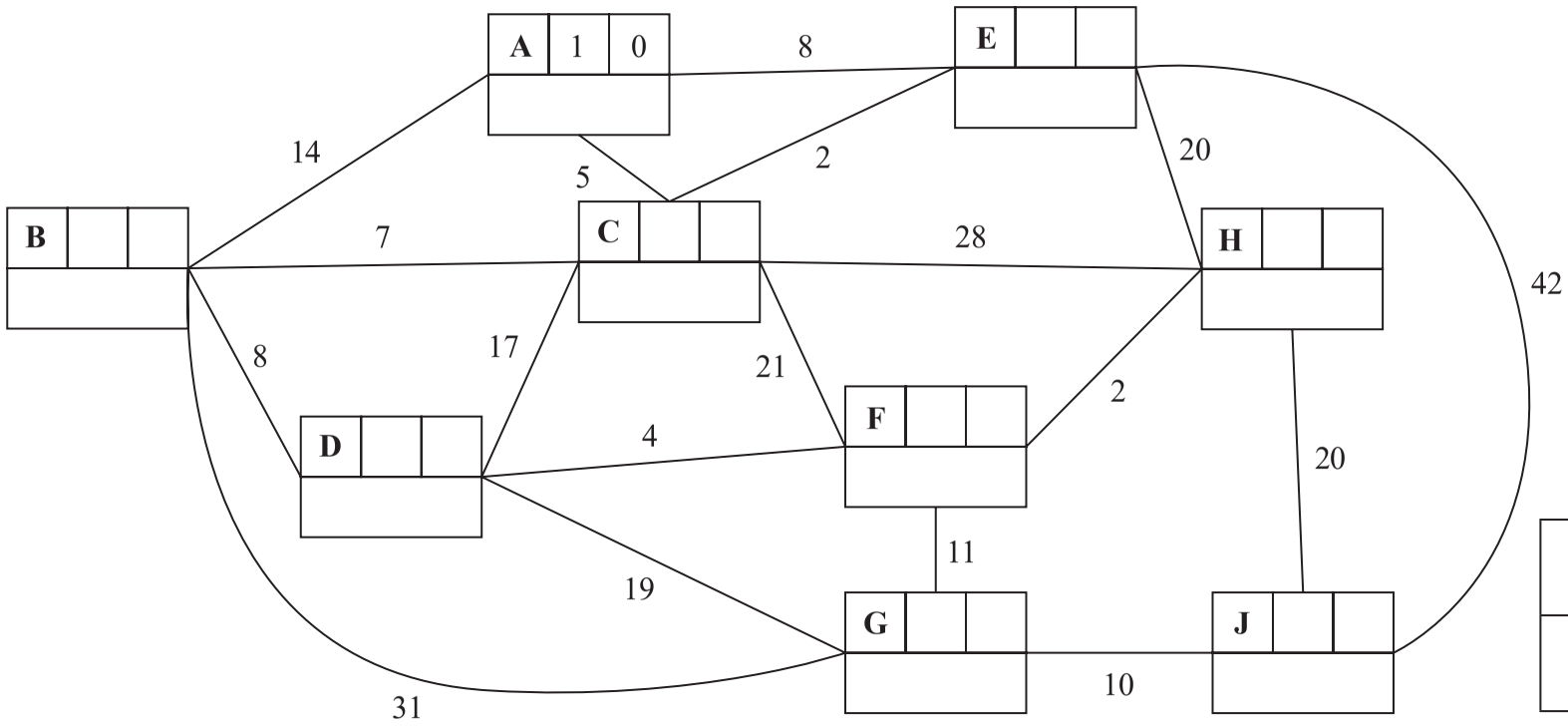
(2)

(d) Determine whether it would be quicker for Alan to start and finish his inspection route at H, instead of starting at H and finishing at D. You must explain your reasoning and show all your working.

(2)

(Total 15 marks)

6.



Key:

Vertex	Order of labelling	Final value
Working values		

Shortest time: _____

Quickest route: _____

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Question 6 continued

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Question 6 continued

Q6

(Total 15 marks)

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4. (a) Explain the difference between the classical and the practical travelling salesperson problems.

(2)

The table below shows the distances, in km, between seven museums, A, B, C, D, E, F and G.

	A	**B**	**C**	**D**	**E**	**F**	**G**
A	–	25	31	28	35	30	32
B	25	–	34	24	27	32	39
C	31	34	–	40	35	27	29
D	28	24	40	–	37	35	36
E	35	27	35	37	–	28	31
F	30	32	27	35	28	–	33
G	32	39	29	36	31	33	–

Fran must visit each museum. She will start and finish at A and wishes to minimise the total distance travelled.

(b) Starting at A, use the nearest neighbour algorithm to obtain an upper bound for the length of Fran's route. Make your method clear.

(2)

Starting at D, a second upper bound of 203 km was found.

(c) State whether this is a better upper bound than the answer to (b), giving a reason for your answer.

(1)

A reduced network is formed by deleting G and all the arcs that are directly joined to G.

(d) (i) Use Prim's algorithm, starting at A, to construct a minimum spanning tree for the reduced network. You must clearly state the order in which you select the arcs of your tree.

(ii) Hence calculate a lower bound for the length of Fran's route.

(4)

By deleting A, a second lower bound was found to be 188 km.

(e) State whether this is a better lower bound than the answer to (d)(ii), giving a reason for your answer.

(1)

(f) Using only the results from (c) and (e), write down the smallest interval that you can be confident contains the length of Fran's optimal route.

(2)

(Total 12 marks)

Leave blank

4. (a) _____

	A	B	C	D	E	F	G
A	–	25	31	28	35	30	32
B	25	–	34	24	27	32	39
C	31	34	–	40	35	27	29
D	28	24	40	–	37	35	36
E	35	27	35	37	–	28	31
F	30	32	27	35	28	–	33
G	32	39	29	36	31	33	–

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Question 4 continued

	A	B	C	D	E	F	G
A	–	25	31	28	35	30	32
B	25	–	34	24	27	32	39
C	31	34	–	40	35	27	29
D	28	24	40	–	37	35	36
E	35	27	35	37	–	28	31
F	30	32	27	35	28	–	33
G	32	39	29	36	31	33	–

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Question 4 continued

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Question 4 continued

(Total 12 marks)

Q4

5.

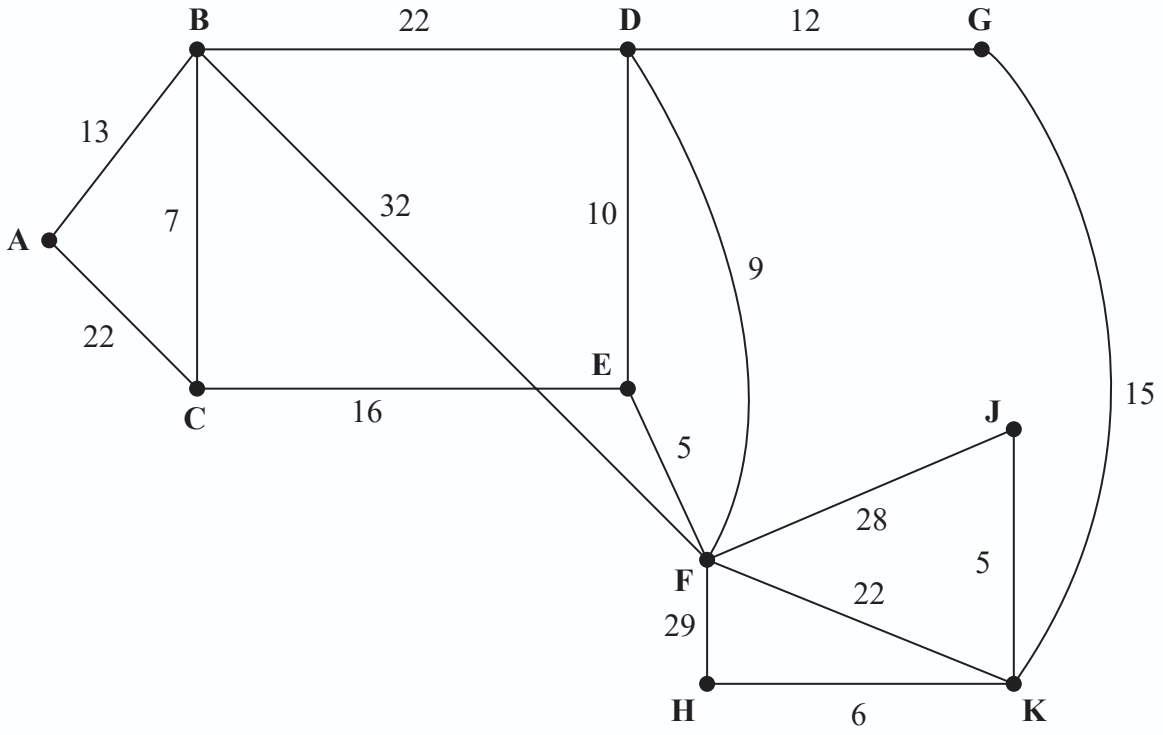


Figure 1

[*The total weight of the network is* 253]

Figure 1 represents a network of roads between 10 cities, A, B, C, D, E, F, G, H, J and K. The number on each edge represents the length, in miles, of the corresponding road.

One day, Mabintou wishes to travel from A to H. She wishes to minimise the distance she travels.

(a) Use Dijkstra's algorithm to find the shortest path from A to H. State your path and its length.

(6)

On another day, Mabintou wishes to travel from F to K via A.

(b) Find a route of minimum length from F to K via A and state its length.

(2)

The roads between the cities need to be inspected. James must travel along each road at least once. He wishes to minimise the length of his inspection route. James will start his inspection route at A and finish at J.

(c) By considering the pairings of all relevant nodes, find the length of James' route. State the arcs that will need to be traversed twice. You must make your method and working clear.

(6)

(d) State the number of times that James will pass through F.

(1)

It is now decided to start the inspection route at D. James must minimise the length of his route. He must travel along each road at least once but may finish at any vertex.

(e) State the vertex where the new inspection route will finish.

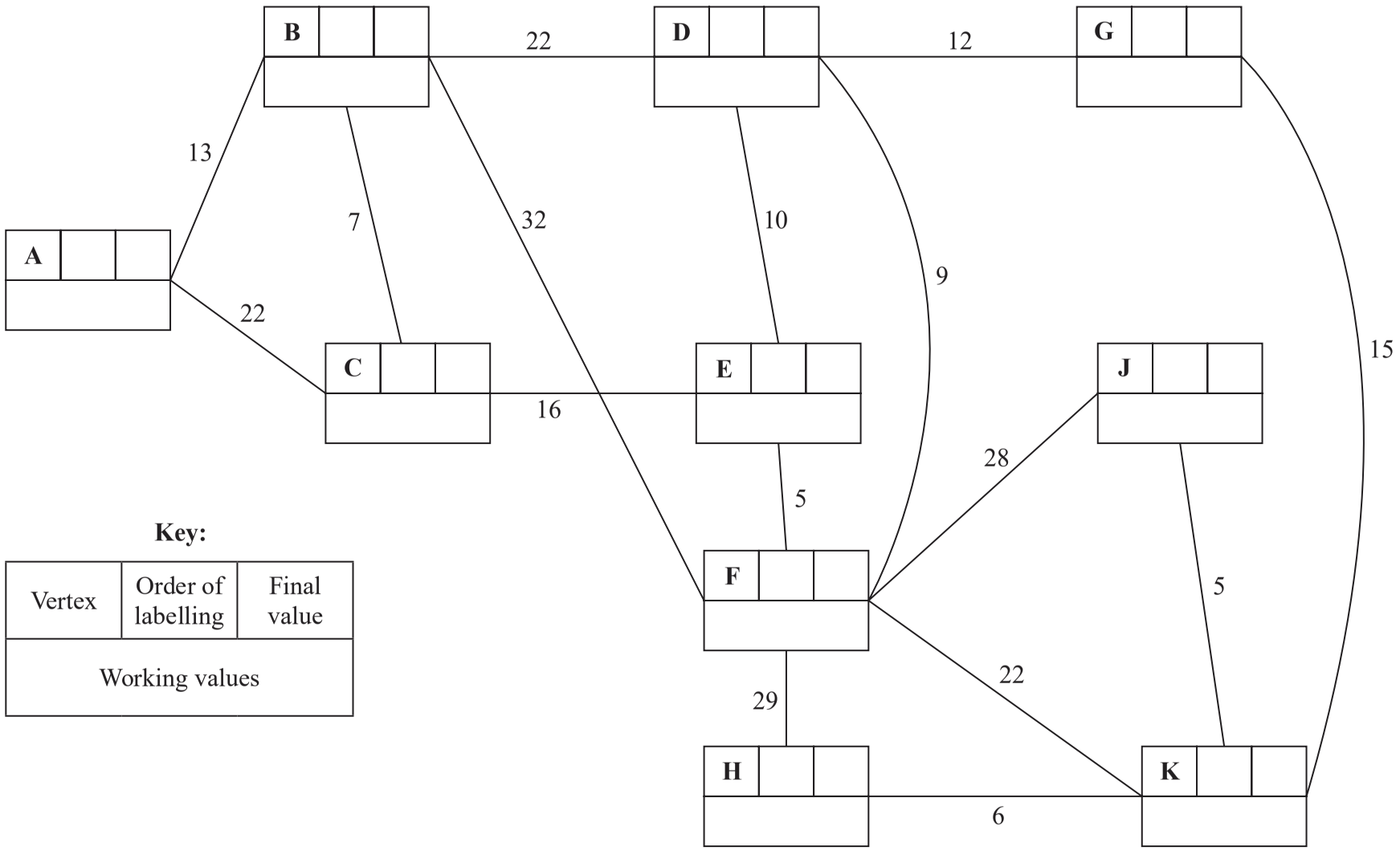
(1)

(f) Calculate the difference between the lengths of the two inspection routes.

(1)

(Total 17 marks)

5.



Key:

Vertex	Order of labelling	Final value
Working values		

Shortest path: _____

Length of shortest path: _____

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Question 5 continued

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Question 5 continued

(Total 17 marks)

Q5

5.

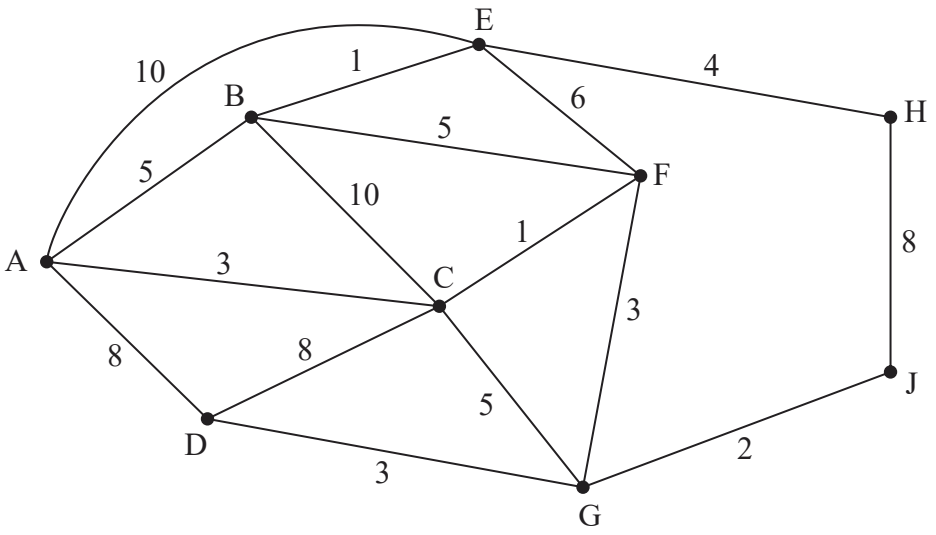


Figure 4

[*The total weight of the network is* 82]

Figure 4 represents a network of 16 roads in a city. The number on each arc represents the time taken, in minutes, to travel along the corresponding road.

Chan needs to check that the roads are in good repair. He must travel along each road at least once. Chan will start and finish at his office at G and must minimise the total time taken for his inspection route.

For this inspection route,

(a) find the time taken and state a possible route. You must make your method and reasoning clear.

(3)

Chan wonders if he can reduce his travel time by starting from his home at B, travelling along each road at least once and finishing at his office at G.

(b) By considering the pairings of all relevant nodes, find any arcs that would need to be traversed twice in the minimum inspection route from B to G. You must make your method clear, showing your working.

(5)

(c) Determine which of the two routes ending at G is quicker, the one starting at G or the one starting at B. You must justify your answer.

(2)

(Total 10 marks)

5.

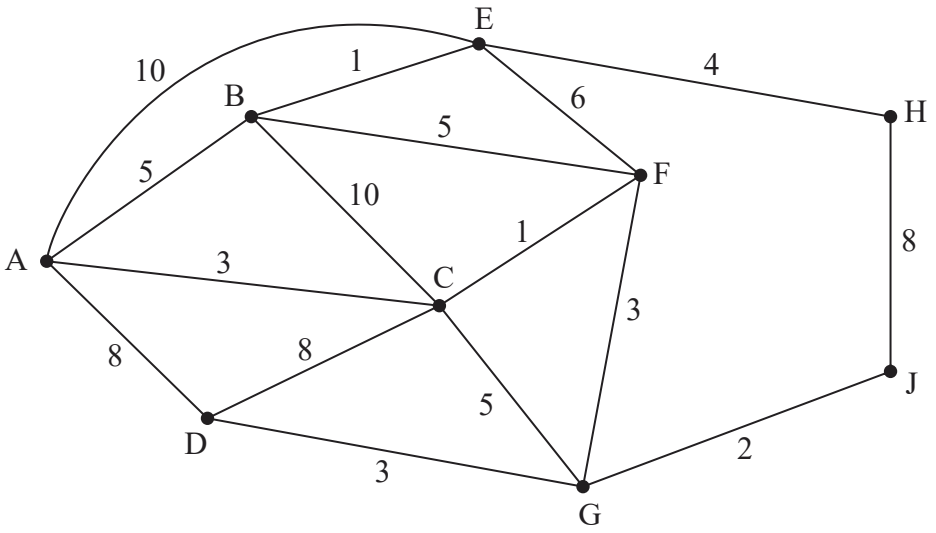


Figure 4

[*The total weight of the network is* 82]

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Question 5 continued

(Total 10 marks)

6.

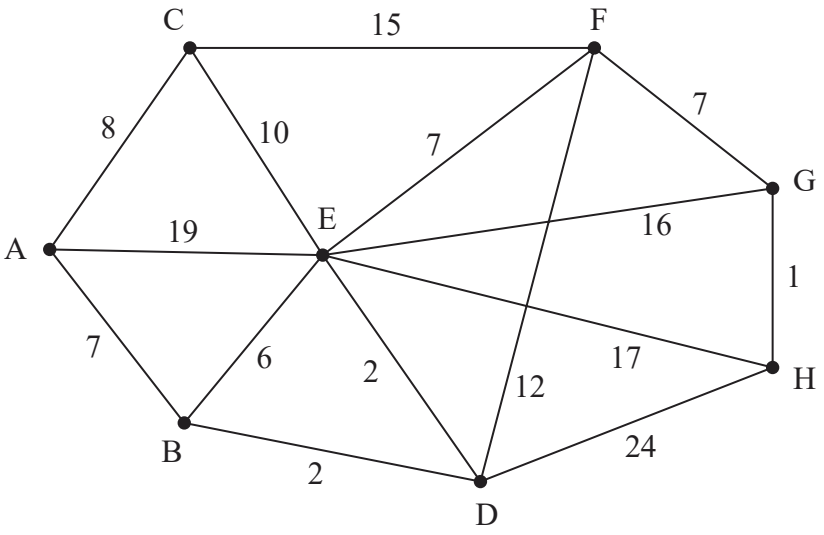


Figure 5

Figure 5 models a network of roads. The number on each edge gives the length, in km, of the corresponding road. The vertices, A, B, C, D, E, F, G and H, represent eight towns. Bronwen needs to visit each town. She will start and finish at A and wishes to minimise the total distance travelled.

(a) By applying Dijkstra's algorithm, starting at A, complete the table of least distances in the answer book.

(6)

(b) Starting at A, use the nearest neighbour algorithm to find an upper bound for the length of Bronwen's route. Write down the route that gives this upper bound.

(2)

A reduced network is formed by deleting A and all arcs that are directly joined to A.

(c) (i) Use Prim's algorithm, starting at C, to construct a minimum spanning tree for the reduced network. You must clearly state the order in which you select the arcs of your tree.

(ii) Hence, calculate a lower bound for the length of Bronwen's route.

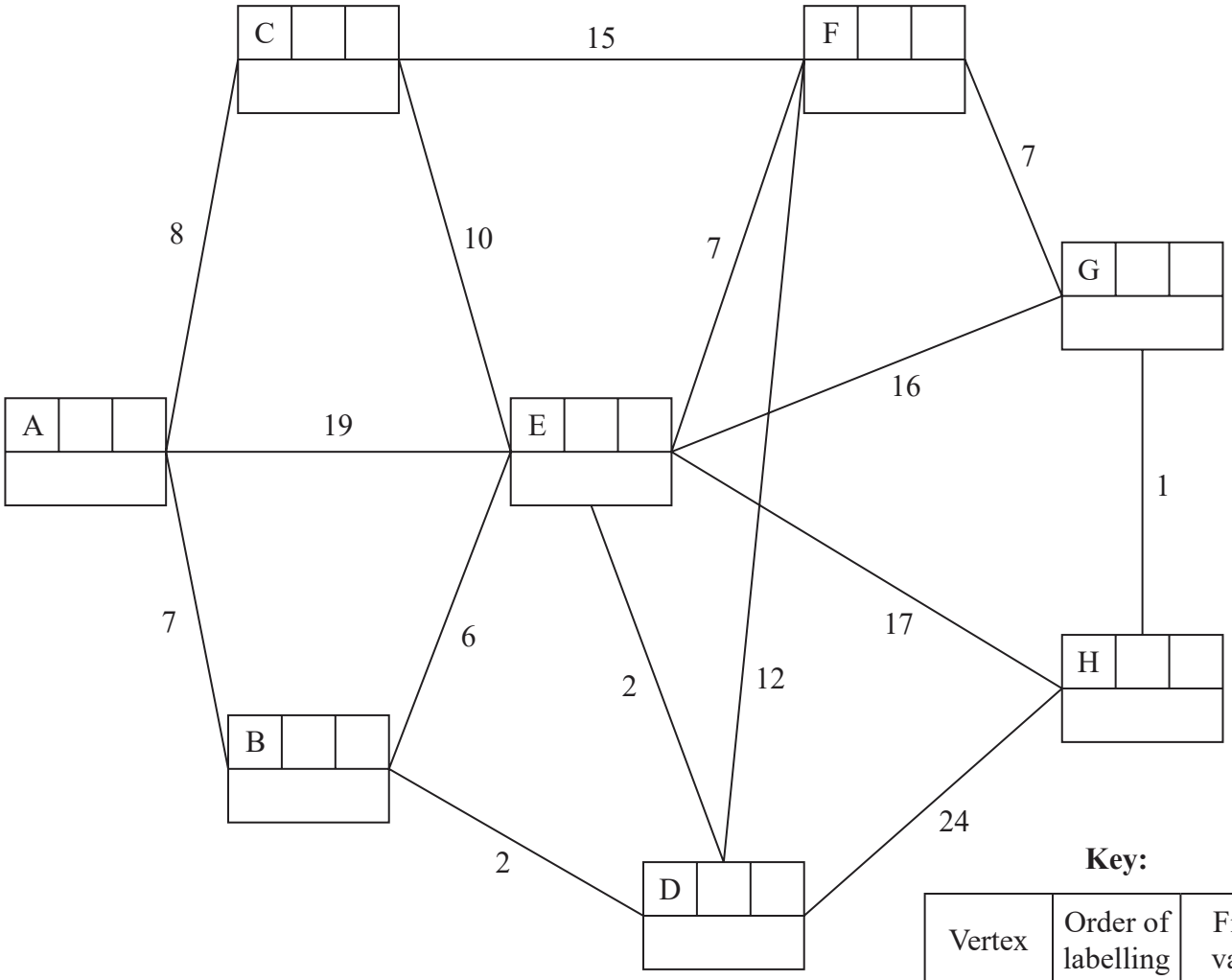
(4)

(d) Using only the results from (b) and (c), write down the smallest interval that you can be confident contains the length of Bronwen's optimal route.

(2)

(Total 14 marks)

6.



	A	B	C	D	E	F	G	H
A	–							
B		–	14	2	4	11	18	19
C		14	–	12	10	15	22	23
D		2	12	–	2	9	16	17
E		4	10	2	–	7	14	15
F		11	15	9	7	–	7	8
G		18	22	16	14	7	–	1
H		19	23	17	15	8	1	–

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Question 6 continued

(b)

(c)

	B	C	D	E	F	G	H
B	–	14	2	4	11	18	19
C	14	–	12	10	15	22	23
D	2	12	–	2	9	16	17
E	4	10	2	–	7	14	15
F	11	15	9	7	–	7	8
G	18	22	16	14	7	–	1
H	19	23	17	15	8	1	–

Question 6 continued

(Total 14 marks)

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Write your answers in the D1 answer book for this paper.

1.

	A	B	C	D	E	F	G
A	–	43	52	47	59	53	55
B	43	–	59	45	46	52	47
C	52	59	–	51	50	55	51
D	47	45	51	–	52	49	55
E	59	46	50	52	–	57	48
F	53	52	55	49	57	–	55
G	55	47	51	55	48	55	–

The table above shows the least distances, in metres, between seven classrooms, A, B, C, D, E, F and G. A teacher needs to visit each classroom, starting and finishing at A, and wishes to minimise the total distance travelled.

(a) Show that there are two nearest neighbour routes that start from A. State these routes and their corresponding lengths.

(3)

(b) Starting by deleting A, and all of its arcs, find a lower bound for the length of the teacher's route.

(3)

(c) Use your results to write down the smallest interval which you can be confident contains the optimal length of the teacher's route.

(1)

(Total for Question 1 is 7 marks)

1.

	A	B	C	D	E	F	G
A	–	43	52	47	59	53	55
B	43	–	59	45	46	52	47
C	52	59	–	51	50	55	51
D	47	45	51	–	52	49	55
E	59	46	50	52	–	57	48
F	53	52	55	49	57	–	55
G	55	47	51	55	48	55	–

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Question 1 continued

	A	B	C	D	E	F	G
A	–	43	52	47	59	53	55
B	43	–	59	45	46	52	47
C	52	59	–	51	50	55	51
D	47	45	51	–	52	49	55
E	59	46	50	52	–	57	48
F	53	52	55	49	57	–	55
G	55	47	51	55	48	55	–

(Total for Question 1 is 7 marks)

2.

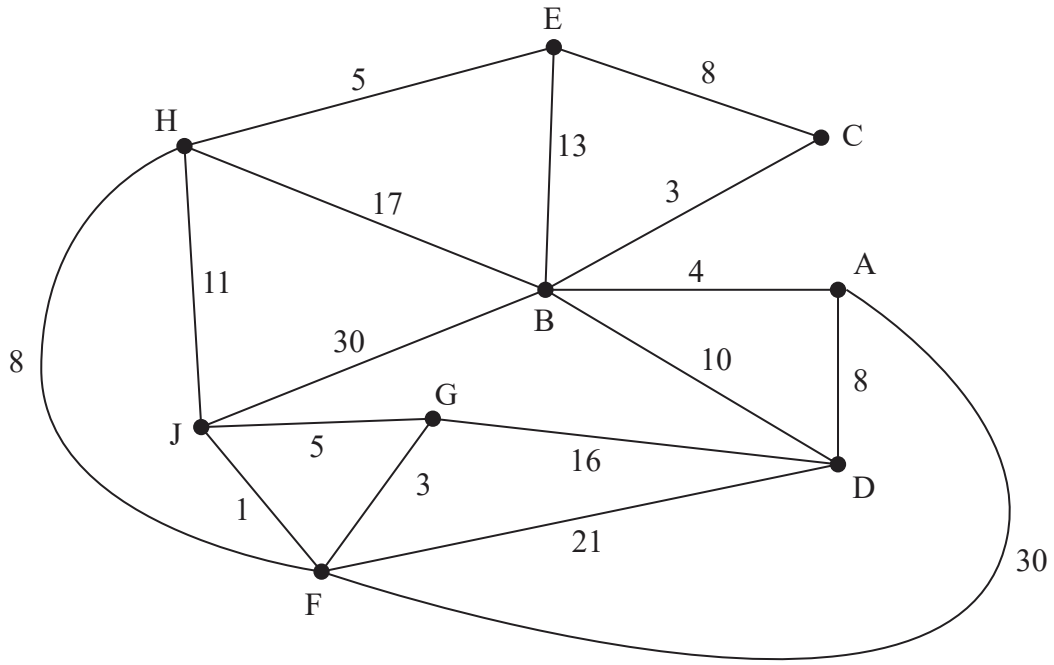


Figure 1

[*The total weight of the network is* 193]

Figure 1 represents a network of roads. The number on each edge represents the length, in miles, of the corresponding road. Jan wishes to travel from A to J. She wishes to minimise the distance she travels.

(a) Use Dijkstra's algorithm to find the shortest path from A to J. Obtain the shortest path and state its length.

(6)

On Monday, Jan needs to travel from her gym at J to her home at H via her office at A.

(b) State the shortest path from J to H via A and its length.

(2)

On Tuesday, Jan needs to check each road. She must travel along each road at least once. Jan must start and finish at A.

(c) Use the route inspection algorithm to find the length of the shortest inspection route. State the roads that should be repeated. You should make your method and working clear.

(5)

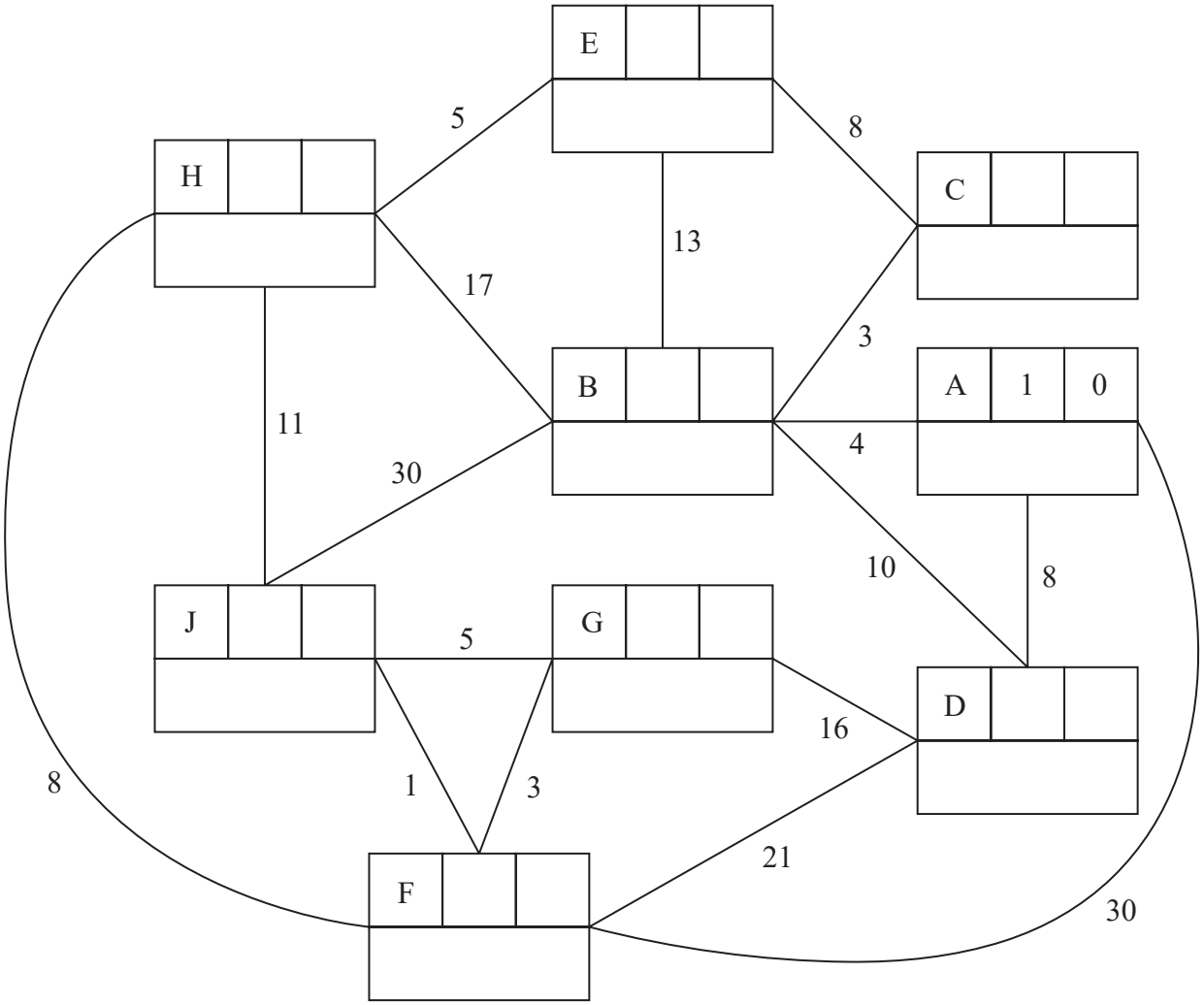
On Wednesday, Jan decides to start her inspection route at G but can finish her route at a different node. The inspection route must still traverse each road at least once.

(d) Determine where the route should finish so that the length of the inspection route is minimised. You must give reasons for your answer and state the length of the route.

(3)

(Total for Question 2 is 16 marks)

2.



Key:

Vertex	Order of labelling	Final value
Working values		

Shortest path from A to J: _____

Length of shortest path from A to J: _____

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Question 2 continued

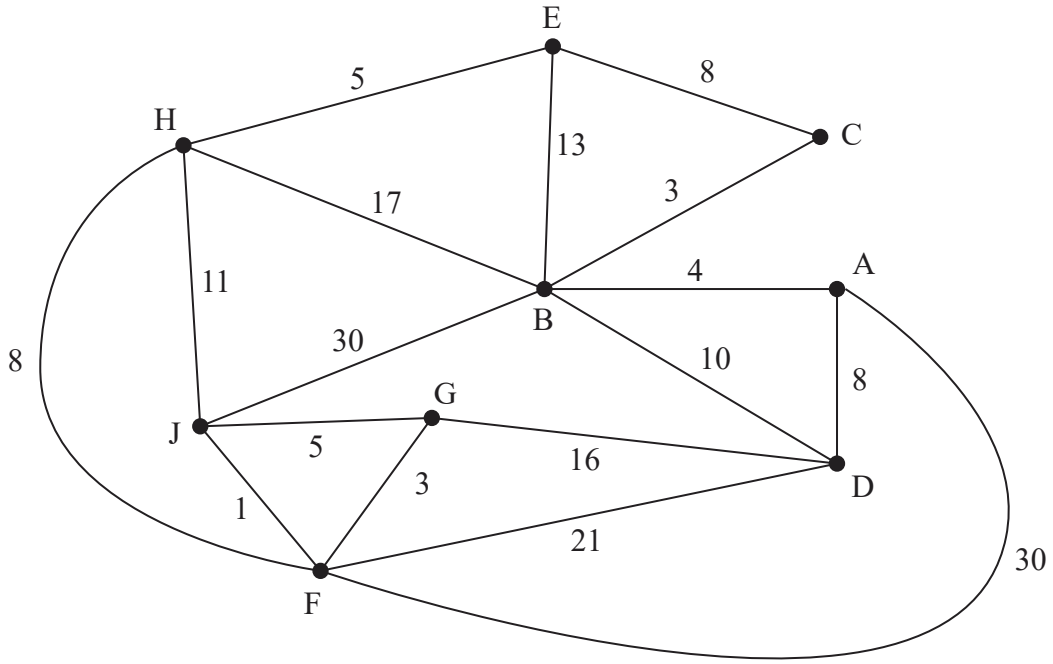


Figure 1

[*The total weight of the network is* 193]

Question 2 continued

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Question 2 continued

(Total for Question 2 is 16 marks)

2.

	A	B	C	D	E	F	G	H
A	–	34	29	35	28	30	37	38
B	34	–	32	28	39	40	32	39
C	29	32	–	27	33	39	34	31
D	35	28	27	–	35	38	41	36
E	28	39	33	35	–	36	33	40
F	30	40	39	38	36	–	34	39
G	37	32	34	41	33	34	–	35
H	38	39	31	36	40	39	35	–

Table 1

Table 1 represents a network that shows the travel times, in minutes, between eight towns, A, B, C, D, E, F, G and H.

(a) Use Prim's algorithm, starting at A, to find the minimum spanning tree for this network. You must clearly state the order in which you select the edges of your tree.

(3)

(b) State the weight of the minimum spanning tree.

(1)

	A	B	C	D	E	F	G	H
J	33	37	41	35	x	40	28	42

Table 2

Table 2 shows the travel times, in minutes, between town J and towns A, B, C, D, E, F, G and H.
The journey time between towns E and J is x minutes where $x > 28$

A salesperson needs to visit all of the **nine** towns, starting and finishing at J.
The salesperson wishes to minimise the total time spent travelling.

(c) Starting at J, use the nearest neighbour algorithm to find an upper bound for the duration of the salesperson's route. Write down the route that gives this upper bound.

(2)

Using the nearest neighbour algorithm, starting at E, an upper bound of 291 minutes for the salesperson's route was found.

(d) State the best upper bound that can be obtained by using this information and your answer to (c). Give the reason for your answer.

(1)

Starting by deleting J and all of its arcs, a lower bound of 264 minutes for the duration of the salesperson's route was found.

(e) Determine the value of x. You must make your method and working clear.

(3)

(Total for Question 2 is 10 marks)

2.

	A	B	C	D	E	F	G	H
A	–	34	29	35	28	30	37	38
B	34	–	32	28	39	40	32	39
C	29	32	–	27	33	39	34	31
D	35	28	27	–	35	38	41	36
E	28	39	33	35	–	36	33	40
F	30	40	39	38	36	–	34	39
G	37	32	34	41	33	34	–	35
H	38	39	31	36	40	39	35	–

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Question 2 continued

	A	B	C	D	E	F	G	H
A	–	34	29	35	28	30	37	38
B	34	–	32	28	39	40	32	39
C	29	32	–	27	33	39	34	31
D	35	28	27	–	35	38	41	36
E	28	39	33	35	–	36	33	40
F	30	40	39	38	36	–	34	39
G	37	32	34	41	33	34	–	35
H	38	39	31	36	40	39	35	–

(Total for Question 2 is 10 marks)

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3.

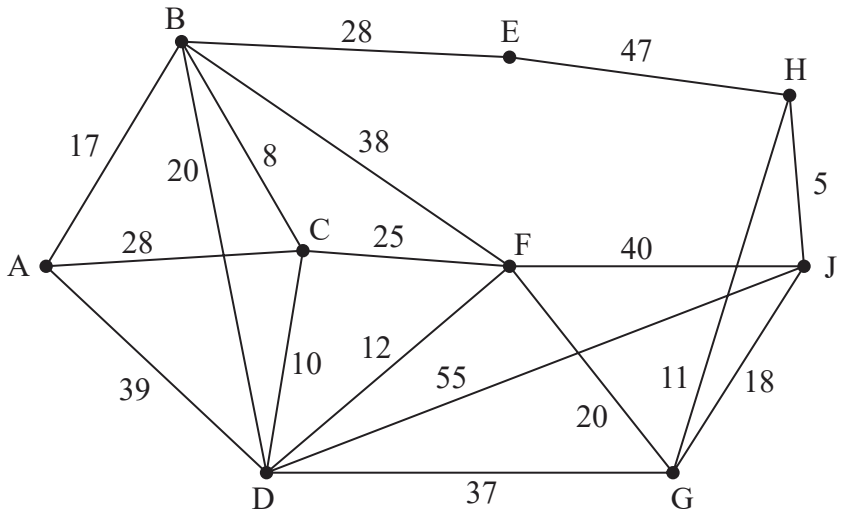


Figure 2

[*The total weight of the network is* 458]

Figure 2 represents a network of roads between nine towns, A, B, C, D, E, F, G, H and J. The number on each edge represents the length, in kilometres, of the corresponding road.

(a) (i) Use Dijkstra's algorithm to find the shortest path from A to J.

(ii) State the length of the shortest path from A to J.

(6)

The roads between the towns must be inspected. Claude must travel along each road at least once. Claude will start the inspection route at A and finish at J. Claude wishes to minimise the length of the inspection route.

(b) By considering the pairings of all relevant nodes, find the length of Claude's route. State the arcs that will need to be traversed twice.

(5)

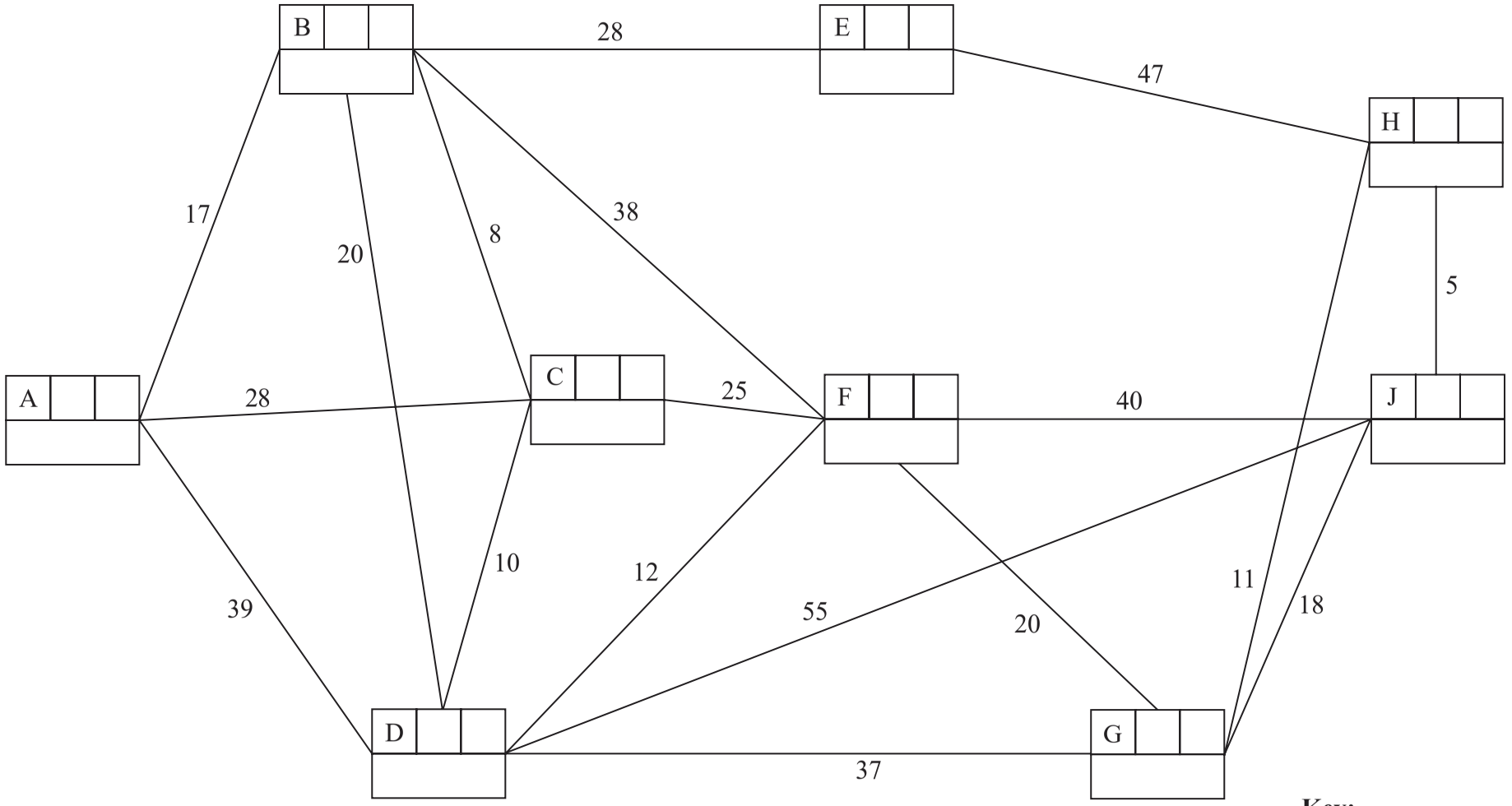
If Claude does **not** start the inspection route at A and finish at J, a shorter inspection route is possible.

(c) Determine the two towns at which Claude should start and finish so that the route has minimum length. Give a reason for your answer and state the length of this route.

(3)

(Total for Question 3 is 14 marks)

3.



Key:

Vertex	Order of labelling	Final value
Working values		

Shortest path from A to J: _____

Length of shortest path from A to J: _____

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Question 3 continued

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Question 3 continued

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Question 3 continued

(Total for Question 3 is 14 marks)

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3. The table below shows the least distances, in km, between six towns, A, B, C, D, E and F.

	A	**B**	**C**	**D**	**E**	**F**
A	–	57	76	59	72	65
B	57	–	67	80	66	76
C	76	67	–	71	83	80
D	59	80	71	–	77	78
E	72	66	83	77	–	69
F	65	76	80	78	69	–

Mei must visit each town at least once. She will start and finish at A and wishes her route to minimise the total distance she will travel.

(a) Starting with the minimum spanning tree in the answer book, use the shortcut method to find an upper bound below 520 km for Mei's route. You must state the shortcut(s) you use and the length of your upper bound.

(2)

(b) Use the nearest neighbour algorithm, starting at A, to find another upper bound for the length of Mei's route.

(2)

(c) Starting by deleting E, and all of its arcs, find a lower bound for the length of Mei's route. Make your method clear.

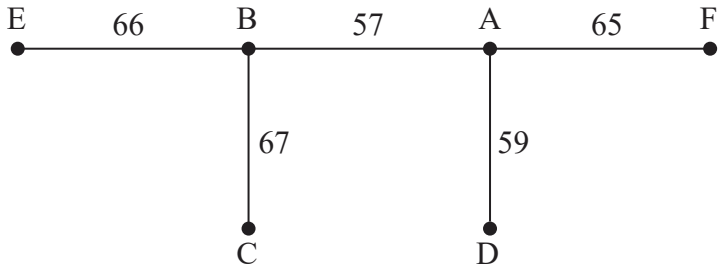
(3)

(Total 7 marks)

Leave blank

3.

	A	B	C	D	E	F
A	–	57	76	59	72	65
B	57	–	67	80	66	76
C	76	67	–	71	83	80
D	59	80	71	–	77	78
E	72	66	83	77	–	69
F	65	76	80	78	69	–



The total weight of the minimum spanning tree is 314 km

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Question 3 continued

	A	B	C	D	E	F
A	–	57	76	59	72	65
B	57	–	67	80	66	76
C	76	67	–	71	83	80
D	59	80	71	–	77	78
E	72	66	83	77	–	69
F	65	76	80	78	69	–

Q3

(Total 7 marks)

7.

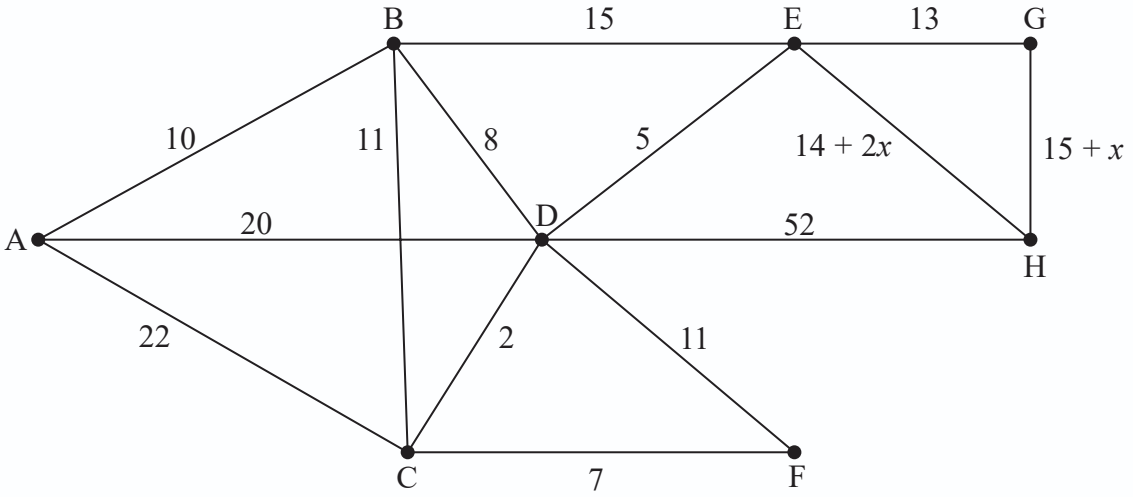


Figure 3

[*The total weight of the network is* $205 + 3x$]

Figure 3 represents a network of roads. The number on each arc represents the time taken, in minutes, to drive along the corresponding road.

Malcolm wishes to minimise the time spent driving from his home at A to his office at H. The delays from roadworks on two of the roads leading in to H vary daily, and so the time taken to drive along these roads is expressed in terms of x, where x is fixed for any given day and $x > 0$

(a) Use Dijkstra's algorithm to find the possible routes that minimise the driving time from A to H. State the length of each route, leaving your answer in terms of x where necessary.

(7)

On Monday, Malcolm needs to check each road. He must travel along each road at least once. He must start and finish at H and minimise the total time taken for his inspection route.

Malcolm finds that his minimum duration inspection route requires him to traverse exactly four roads twice and the total time it takes to complete his inspection route is 307 minutes.

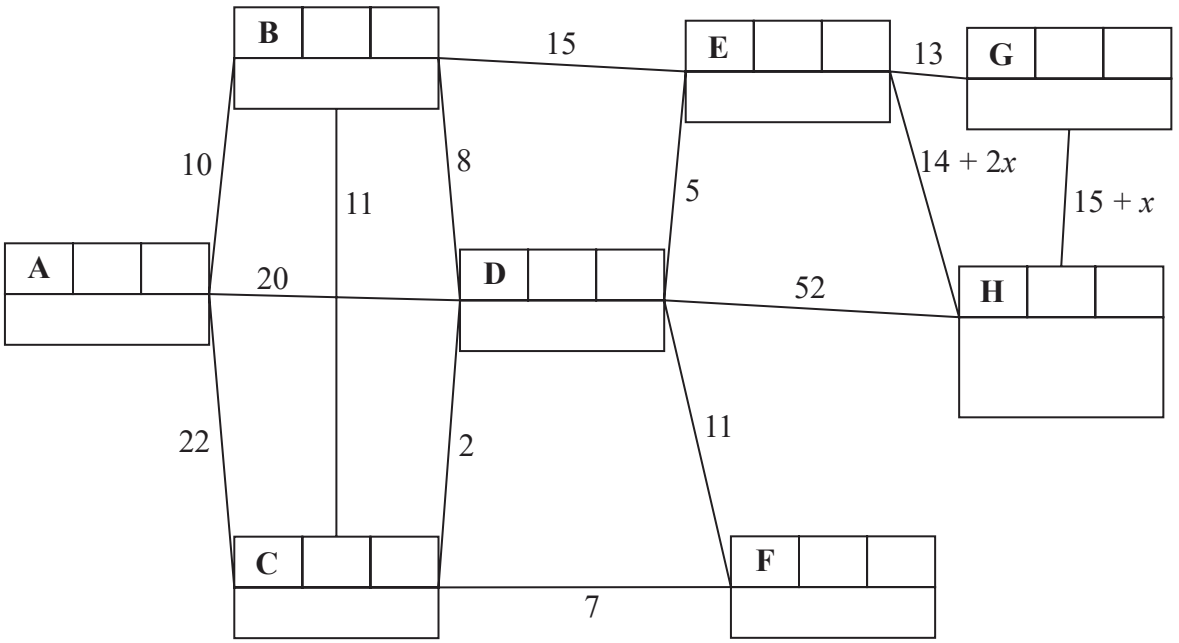
(b) Calculate the minimum time taken for Malcolm to travel from A to H on Monday. You must make your method and working clear.

(4)

(Total 11 marks)

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7.



Key:

Vertex	Order of labelling	Final value
Working values		

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Question 7 continued

(Total 11 marks)

Q7

4.

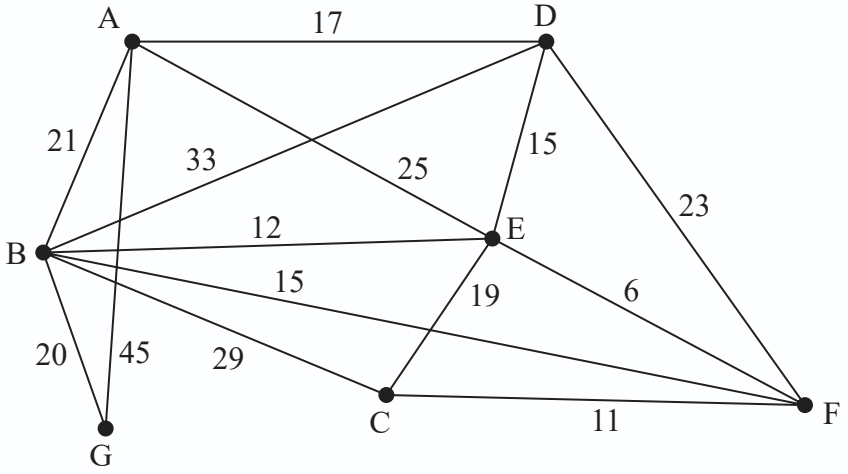


Figure 3

[*The total weight of the network is* 291]

Figure 3 models a network of roads. The number on each edge gives the length, in km, of the corresponding road. The vertices, A, B, C, D, E, F and G, represent seven towns. Derek needs to visit each town. He will start and finish at A and wishes to minimise the total distance travelled.

(a) By inspection, complete the two copies of the table of least distances in the answer book.

(2)

(b) Starting at A, use the nearest neighbour algorithm to find an upper bound for the length of Derek's route. Write down the route that gives this upper bound.

(2)

(c) Interpret the route found in (b) in terms of the towns actually visited.

(1)

(d) Starting by deleting A and all of its arcs, find a lower bound for the route length.

(3)

Clive needs to travel along the roads to check that they are in good repair. He wishes to minimise the total distance travelled and must start at A and finish at G.

(e) By considering the pairings of all relevant nodes, find the length of Clive's route. State the edges that need to be traversed twice. You must make your method and working clear.

(5)

(Total 13 marks)

Leave blank

4.

	A	B	C	D	E	F	G
A	–	21		17	25	31	41
B	21	–	26	27	12	15	20
C		26	–		17	11	46
D	17	27		–	15		47
E	25	12	17	15	–	6	32
F	31	15	11		6	–	35
G	41	20	46	47	32	35	–

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Question 4 continued

	A	B	C	D	E	F	G
A	–	21		17	25	31	41
B	21	–	26	27	12	15	20
C		26	–		17	11	46
D	17	27		–	15		47
E	25	12	17	15	–	6	32
F	31	15	11		6	–	35
G	41	20	46	47	32	35	–

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Question 4 continued

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Question 4 continued

(Total 13 marks)

Q4

5.

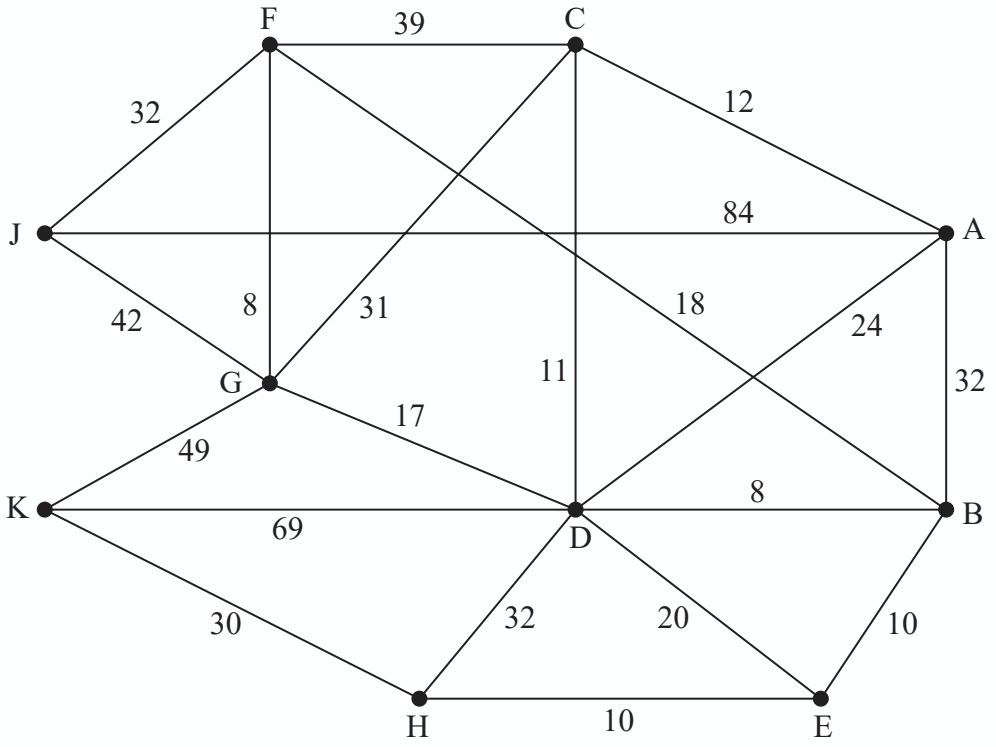


Figure 4

Figure 4 represents a network of roads. The number on each arc represents the length, in miles, of the corresponding road. Tamasi, who lives at A, needs to collect a caravan. Tamasi can collect a caravan from either J or K.

Tamasi decides to use Dijkstra's algorithm once to find the shortest routes between A and J and between A and K.

(a) State, with a reason, which vertex should be chosen as the starting vertex for the algorithm.

(2)

(b) Use Dijkstra's algorithm to find the shortest routes from A to J and from A to K. You should state the routes and their corresponding lengths.

(7)

Tamasi's brother lives at F. He needs to visit Tamasi at A and then visit their mother who lives at H.

(c) Find a route of minimal length that goes from F to H via A.

(1)

(Total 10 marks)

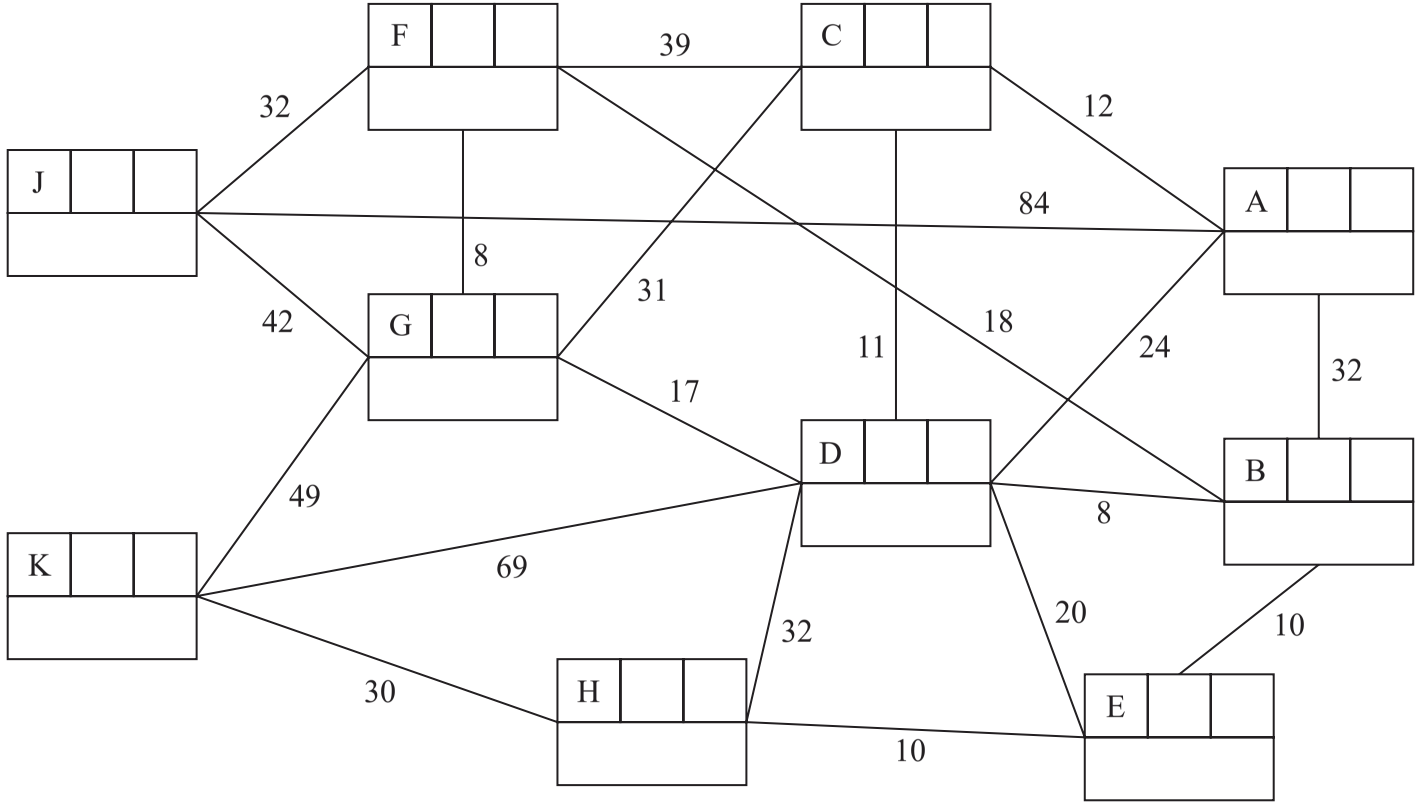
5.

(a) _____

_____

_____

_____



Key:

Vertex	Order of labelling	Final value
Working values		

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Question 5 continued

(Total 10 marks)

Q5

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6.

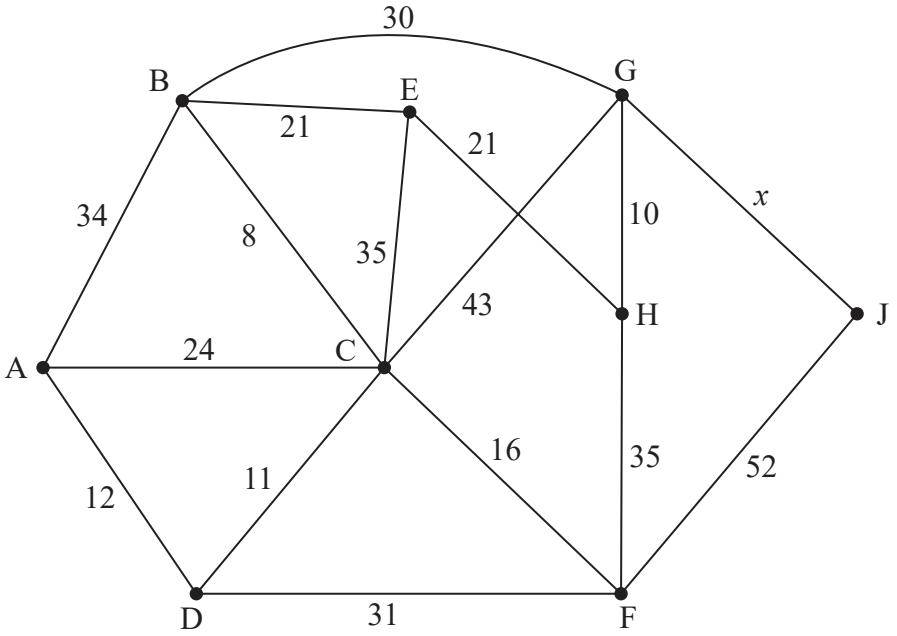


Figure 4

[*The total weight of the network is* 383 + *x*]

Figure 4 models a network of roads. The number on each edge gives the time, in minutes, to travel along the corresponding road. The vertices, A, B, C, D, E, F, G, H and J represent nine towns. Ezra wishes to travel from A to H as fast as possible.

The time taken to travel between towns G and J is unknown and is denoted by x minutes.

Dijkstra's algorithm is to be used to find the fastest time to travel from A to H. On Diagram 1 in the answer book the "Order of labelling" and "Final value" at A and J, and the "Working values" at J, have already been completed.

(a) Use Dijkstra's algorithm to find the fastest time to travel from A to H. State the quickest route.

(6)

Ezra needs to travel along each road to check it is in good repair. He wishes to minimise the total time required to traverse the network. Ezra plans to start and finish his inspection route at A. It is given that his route will take at least 440 minutes.

(b) Use the route inspection algorithm and the completed Diagram 1 to find the range of possible values of x.

(6)

(c) Write down a possible route for Ezra.

(1)

A new direct road from D to H is under construction and will take 25 minutes to travel along. Ezra will include this new road in a minimum length inspection route starting and finishing at A. It is given that this inspection route takes exactly 488 minutes.

(d) Determine the value of x. You must give reasons for your answer.

(2)

(Total for Question 6 is 15 marks)

6.

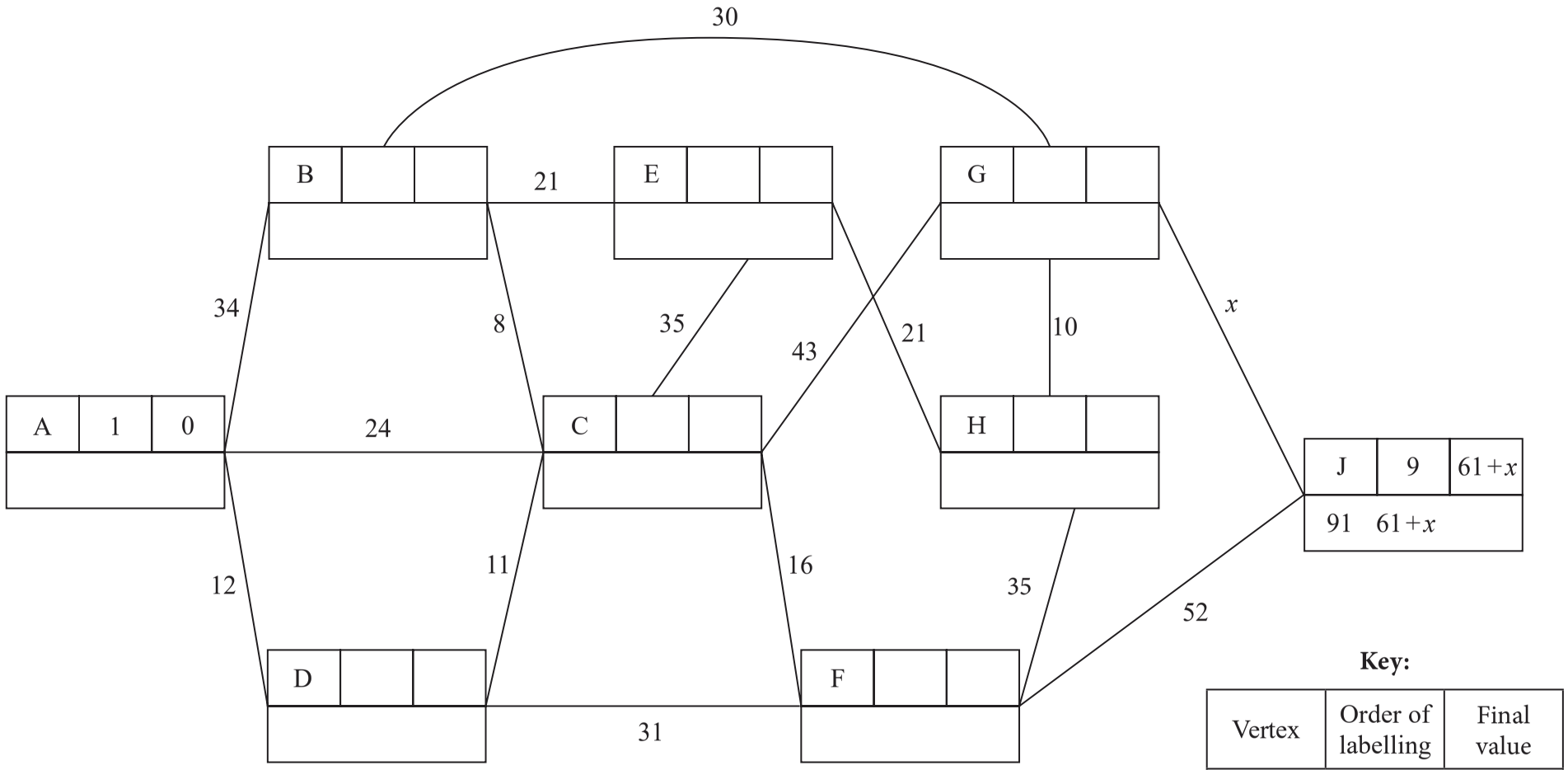


Diagram 1

Fastest time: _____

Quickest route: _____

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Question 6 continued

Question 6 continued

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Question 6 continued

(Total for Question 6 is 15 marks)

6.

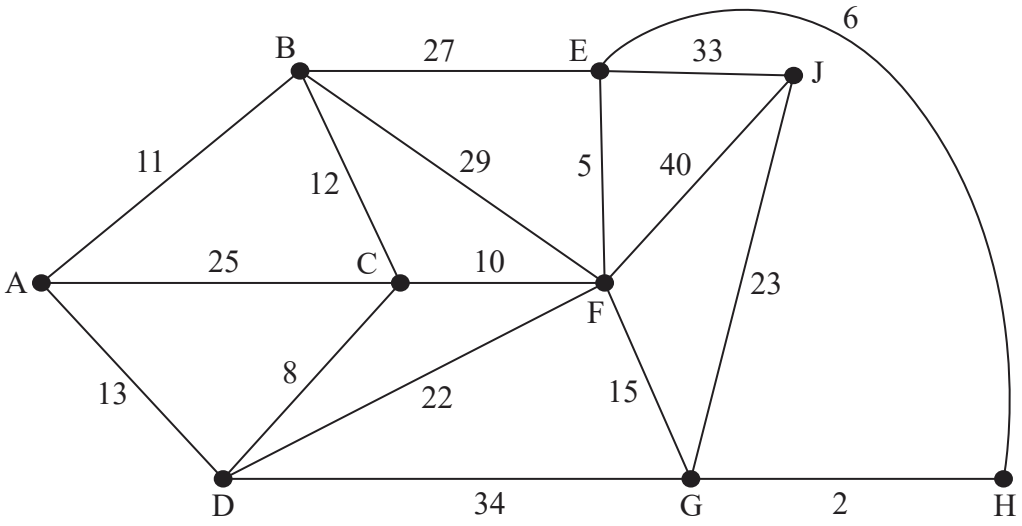


Figure 3

[*The total weight of the network is* 315]

Figure 3 represents a network of roads between nine parks, A, B, C, D, E, F, G, H and J. The number on each edge represents the length, in miles, of the corresponding road.

(a) (i) Use Dijkstra's algorithm to find the shortest path from A to J.

(ii) State the length of the shortest path from A to J.

(6)

The roads between the parks need to be inspected. Robin must travel along each road at least once. Robin wishes to minimise the length of the inspection route. Robin will start the inspection route at C and finish at E.

(b) By considering the pairings of all relevant nodes, find the length of Robin's route.

(4)

(c) State the number of times Robin will pass through G.

(1)

It is now decided to start and finish the inspection route at A. Robin must still minimise the length of the route and travel along each road at least once.

(d) Calculate the difference between the lengths of the two inspection routes.

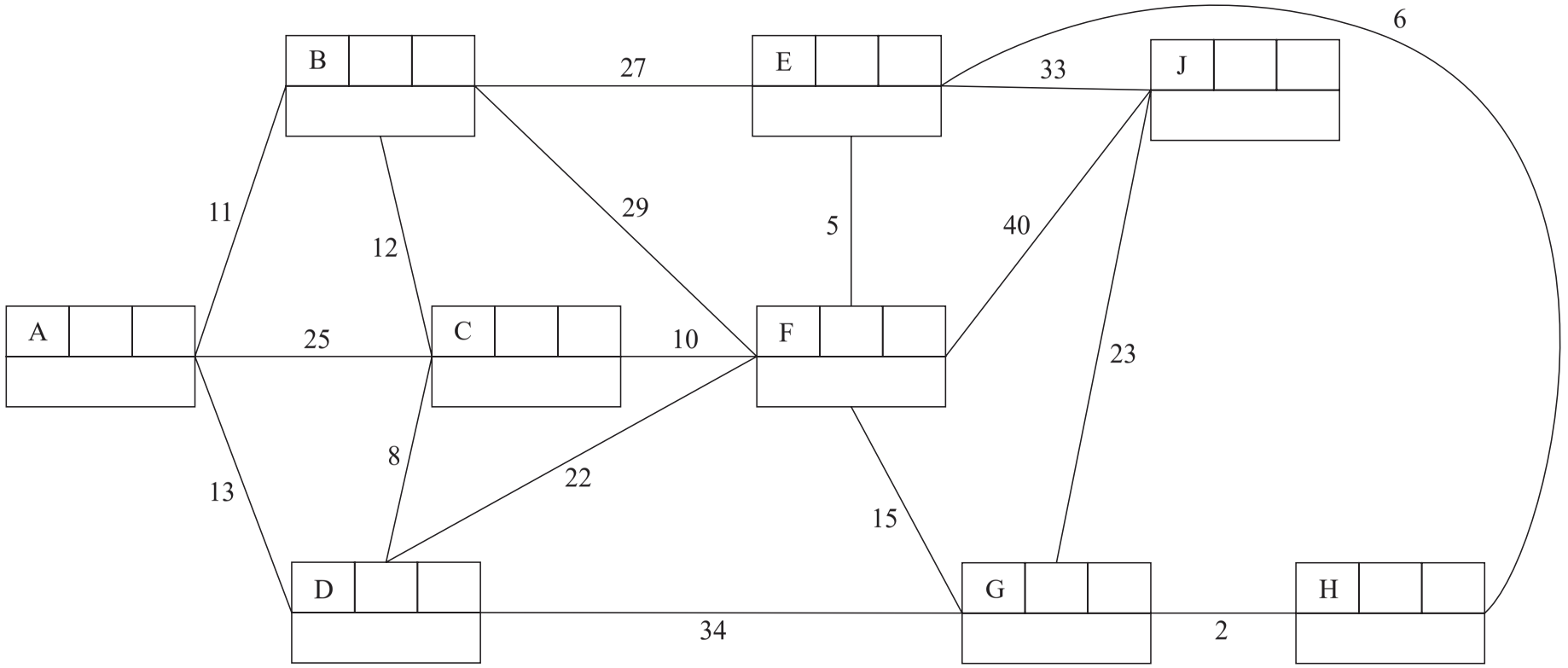
(1)

(e) State the edges that need to be traversed twice in the route that starts and finishes at A, but do not need to be traversed twice in the route that starts at C and finishes at E.

(1)

(Total for Question 6 is 13 marks)

6.



Key:

Vertex	Order of labelling	Final value
Working values		

Shortest path: _____

Length of shortest path: _____

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Question 6 continued

Question 6 continued

(Total for Question 6 is 13 marks)

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7.

	A	B	C	D	E	F	G	H
A	–	38	37	x	37	42	41	27
B	38	–	26	32	33	38	37	34
C	37	26	–	39	38	39	30	39
D	x	32	39	–	37	36	29	36
E	37	33	38	37	–	32	33	30
F	42	38	39	36	32	–	31	28
G	41	37	30	29	33	31	–	33
H	27	34	39	36	30	28	33	–

The network represented by the table shows the least distances, in km, between eight museums, A, B, C, D, E, F, G and H.

A tourist wants to visit each museum at least once, starting and finishing at A. The tourist wishes to minimise the total distance travelled. The shortest distance between A and D is x km where $32 \leqslant x \leqslant 35$

(a) Using Prim's algorithm, starting at A, obtain a minimum spanning tree for the network. You must clearly state the order in which you select the arcs of your tree.

(3)

(b) Use your answer to (a) to determine an initial upper bound for the length of the tourist's route.

(1)

(c) Starting at A, use the nearest neighbour algorithm to find another upper bound for the length of the tourist's route. Write down the route that gives this upper bound.

(2)

The nearest neighbour algorithm starting at E gives a route of

E – H – A – D – G – C – B – F – E

(d) State which of these two nearest neighbour routes gives the better upper bound. Give reasons for your answer.

(2)

Starting by deleting A, and all of its arcs, a lower bound of 235 km for the length of the route is found.

(e) Determine the smallest interval that must contain the optimal length of the tourist's route. You must make your method and working clear.

(4)

(Total for Question 7 is 12 marks)

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7.

	A	B	C	D	E	F	G	H
A	–	38	37	x	37	42	41	27
B	38	–	26	32	33	38	37	34
C	37	26	–	39	38	39	30	39
D	x	32	39	–	37	36	29	36
E	37	33	38	37	–	32	33	30
F	42	38	39	36	32	–	31	28
G	41	37	30	29	33	31	–	33
H	27	34	39	36	30	28	33	–

Question 7 continued

	A	B	C	D	E	F	G	H
A	–	38	37	x	37	42	41	27
B	38	–	26	32	33	38	37	34
C	37	26	–	39	38	39	30	39
D	x	32	39	–	37	36	29	36
E	37	33	38	37	–	32	33	30
F	42	38	39	36	32	–	31	28
G	41	37	30	29	33	31	–	33
H	27	34	39	36	30	28	33	–

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Question 7 continued

	A	B	C	D	E	F	G	H
A	–	38	37	x	37	42	41	27
B	38	–	26	32	33	38	37	34
C	37	26	–	39	38	39	30	39
D	x	32	39	–	37	36	29	36
E	37	33	38	37	–	32	33	30
F	42	38	39	36	32	–	31	28
G	41	37	30	29	33	31	–	33
H	27	34	39	36	30	28	33	–

(Total for Question 7 is 12 marks)

3.

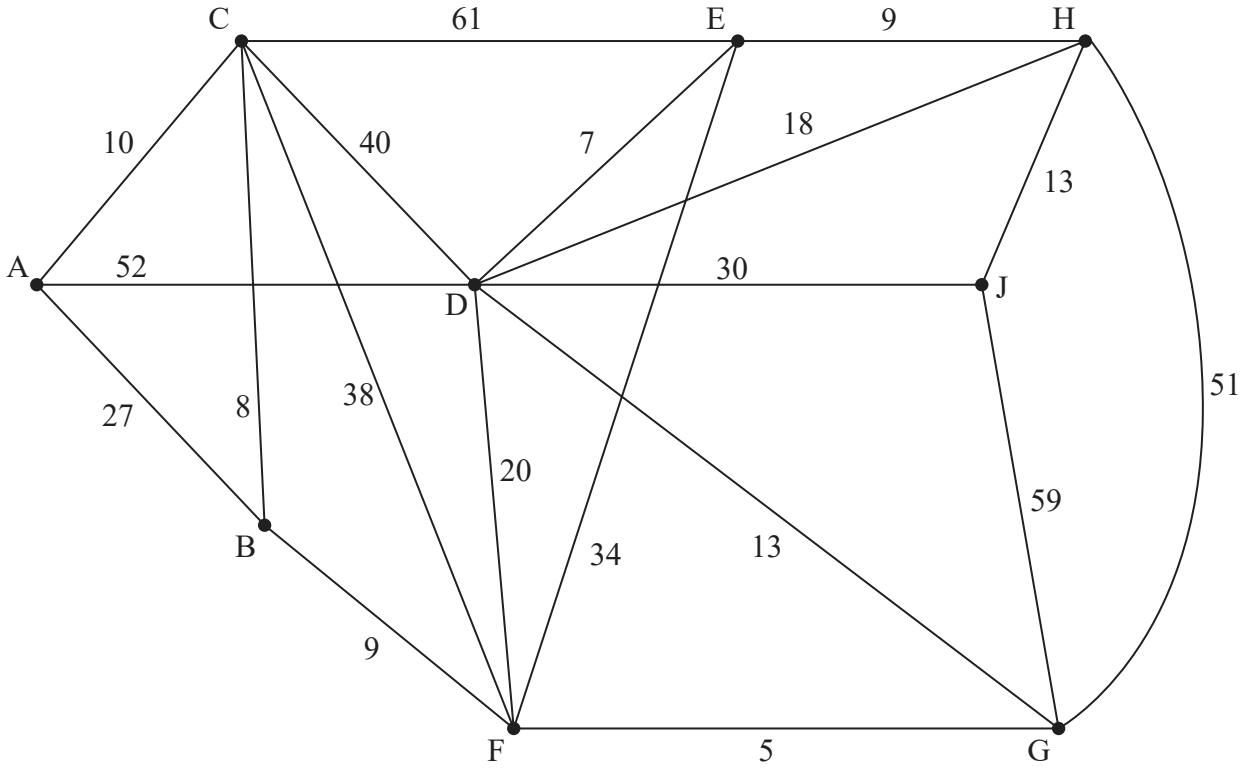


Figure 2

Figure 2 models a network of tracks between nine ranger stations, A, B, C, D, E, F, G, H and J, in a forest. The number on each edge gives the time, in minutes, to travel along the corresponding track. The forest ranger wishes to travel from A to J as quickly as possible.

(a) Use Dijkstra's algorithm to find the shortest time needed to travel from A to J.

State the quickest route.

(6)

(b) Hence determine the weight of the minimum spanning tree for the network given in Figure 2. Give a reason for your answer.

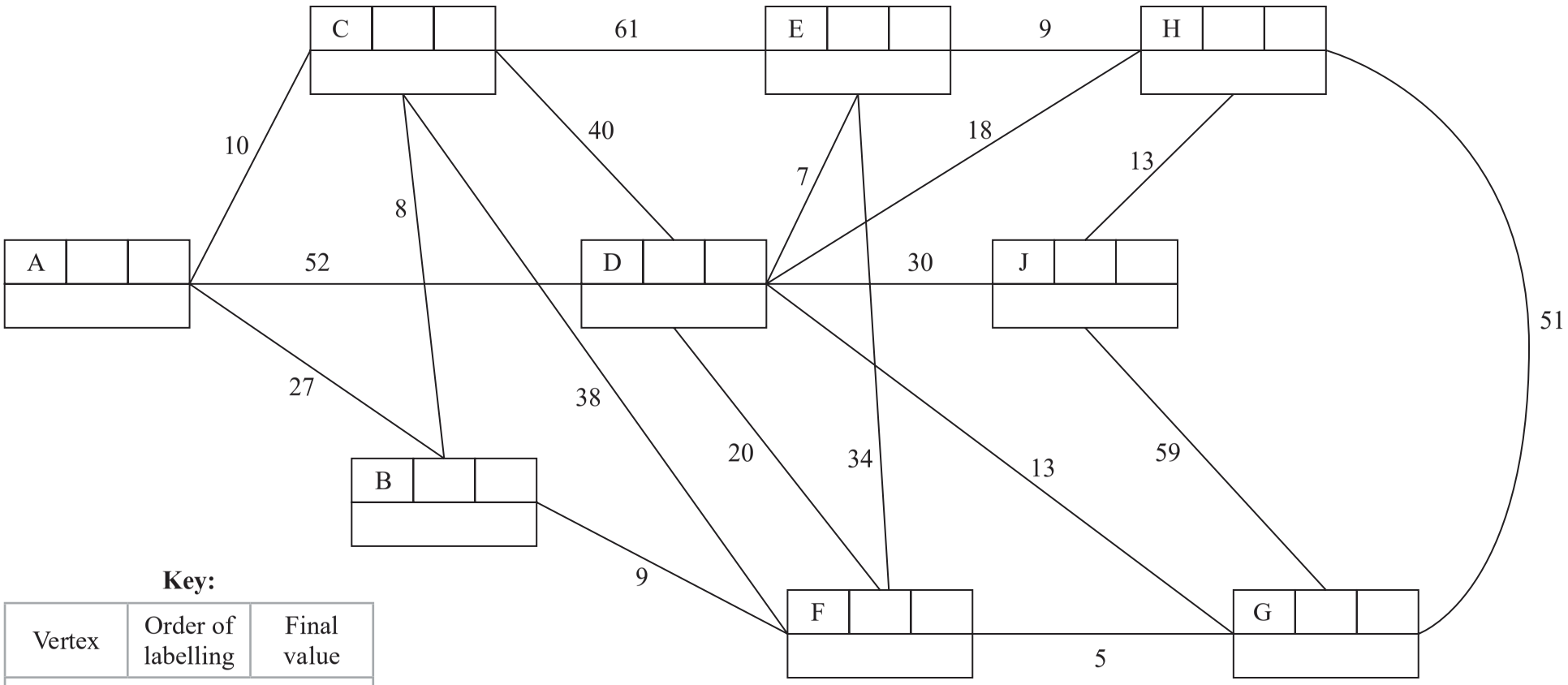
You do not need to find the minimum spanning tree.

(2)

(Total for Question 3 is 8 marks)

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3.



Key:

Vertex	Order of labelling	Final value
Working values		

Shortest time to travel from A to J: _____

Quickest route from A to J: _____

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Question 3 continued

(Total for Question 3 is 8 marks)

3. The table below represents a complete network that shows the least costs of travelling between eight cities, A, B, C, D, E, F, G and H.

	A	B	C	D	E	F	G	H
A	–	36	38	40	23	39	38	35
B	36	–	35	36	35	34	41	38
C	38	35	–	39	25	32	40	40
D	40	36	39	–	37	37	26	33
E	23	35	25	37	–	42	24	43
F	39	34	32	37	42	–	45	38
G	38	41	40	26	24	45	–	40
H	35	38	40	33	43	38	40	–

Srinjoy must visit each city at least once. He will start and finish at A and wishes to minimise his total cost.

(a) Use Prim's algorithm, starting at A, to find a minimum spanning tree for this network. You must list the arcs that form the tree in the order in which you select them.

(3)

(b) State the weight of the minimum spanning tree.

(1)

(c) Use your answer to (b) to help you calculate an initial upper bound for the total cost of Srinjoy's route.

(1)

(d) Show that there are two nearest neighbour routes that start from A. You must make the routes and their corresponding costs clear.

(4)

(e) State the best upper bound that can be obtained by using your answers to (c) and (d).

(1)

(f) Starting by deleting A and all of its arcs, find a lower bound for the total cost of Srinjoy's route. You must make your method and working clear.

(3)

(g) Use your results to write down the smallest interval that must contain the optimal cost of Srinjoy's route.

(2)

(Total 15 marks)

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3.

	A	B	C	D	E	F	G	H
A	–	36	38	40	23	39	38	35
B	36	–	35	36	35	34	41	38
C	38	35	–	39	25	32	40	40
D	40	36	39	–	37	37	26	33
E	23	35	25	37	–	42	24	43
F	39	34	32	37	42	–	45	38
G	38	41	40	26	24	45	–	40
H	35	38	40	33	43	38	40	–

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Question 3 continued

	A	B	C	D	E	F	G	H
A	–	36	38	40	23	39	38	35
B	36	–	35	36	35	34	41	38
C	38	35	–	39	25	32	40	40
D	40	36	39	–	37	37	26	33
E	23	35	25	37	–	42	24	43
F	39	34	32	37	42	–	45	38
G	38	41	40	26	24	45	–	40
H	35	38	40	33	43	38	40	–

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DO NOT WRITE IN THIS AREA

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Question 3 continued

	A	B	C	D	E	F	G	H
A	–	36	38	40	23	39	38	35
B	36	–	35	36	35	34	41	38
C	38	35	–	39	25	32	40	40
D	40	36	39	–	37	37	26	33
E	23	35	25	37	–	42	24	43
F	39	34	32	37	42	–	45	38
G	38	41	40	26	24	45	–	40
H	35	38	40	33	43	38	40	–

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Question 3 continued

	A	B	C	D	E	F	G	H
A	–	36	38	40	23	39	38	35
B	36	–	35	36	35	34	41	38
C	38	35	–	39	25	32	40	40
D	40	36	39	–	37	37	26	33
E	23	35	25	37	–	42	24	43
F	39	34	32	37	42	–	45	38
G	38	41	40	26	24	45	–	40
H	35	38	40	33	43	38	40	–

Q3

(Total 15 marks)

5.

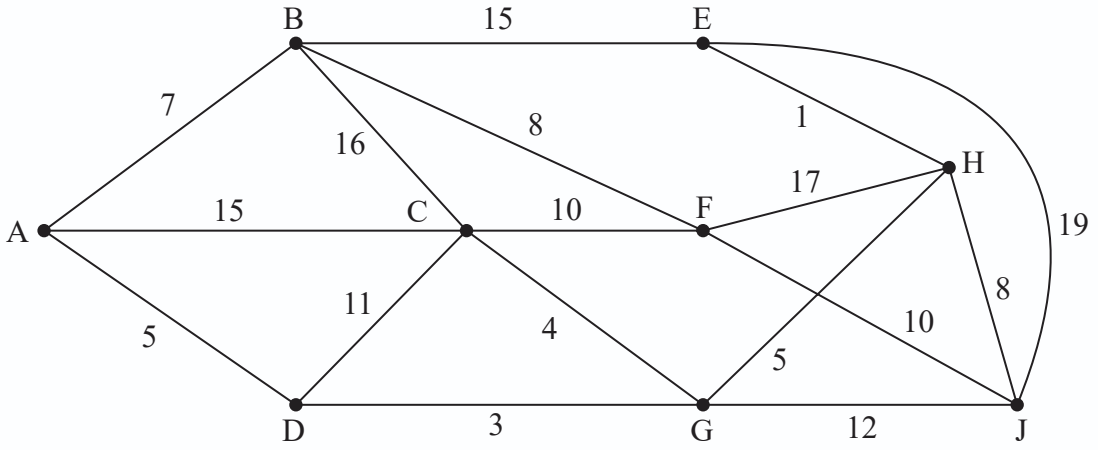


Figure 3

[*The total weight of the network is* 166]

Figure 3 models a network of cycle lanes that must be inspected. The number on each arc represents the length, in km, of the corresponding cycle lane. Lance needs to cycle along each lane at least once and wishes to minimise the length of his inspection route.

He must start and finish at A.

(a) Use an appropriate algorithm to find the length of the route. State the cycle lanes that Lance will need to traverse twice. You should make your method and working clear.

(6)

(b) State the number of times that vertex C appears in Lance's route.

(1)

It is now decided that the inspection route may finish at any vertex. Lance will still start at A and must cycle along each lane at least once.

(c) Determine the finishing point so that the length of the route is minimised. You must give reasons for your answer and state the length of this new minimum route.

(3)

(Total 10 marks)

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5.

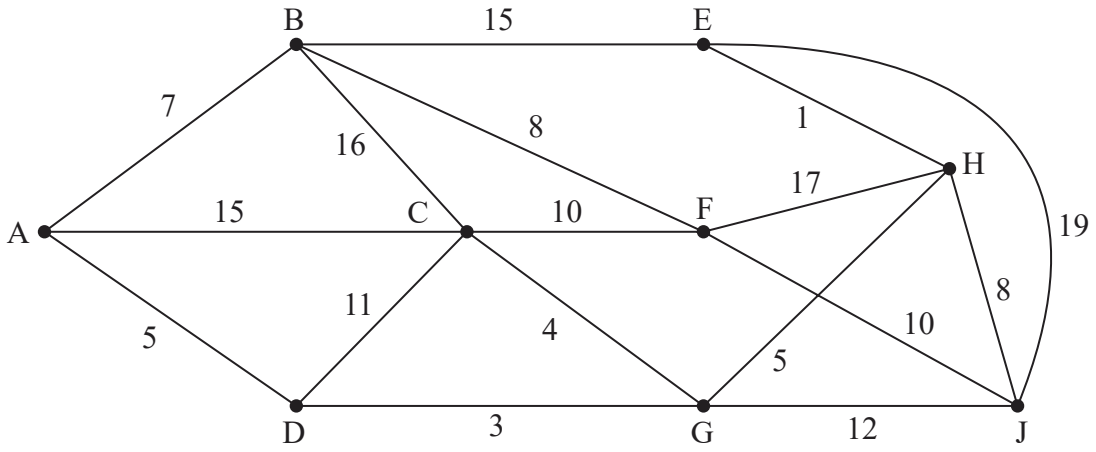


Figure 3

[*The total weight of the network is* 166]

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Question 5 continued

(Total 10 marks)

Q5